Nested Loops & Conditionals

Module 1: Python Foundations

```
for row in range(2):
    for col in range(3):
        print(row, col)
```

By the End of Today, You'll Be Able To…

1

Trace a nested loop's iteration count

2

Trace which nested branch actually runs

CHECK FOR UNDERSTANDING

How many times does `for i in range(4): print(i)` print?

A 3

B 5

C 4

ANSWER KEY

Prior-Lesson Loop Recall

Correct answer: 4

`range(4)` produces 0, 1, 2, 3 — four values. Keep that boundary clear before adding a second loop.

When That’s Not Enough

A grid has rows and columns, not just one list. One loop alone can’t reach every cell.

A Nested Loop Finishes the Inner Loop Every Outer Pass

OUTER

Choose one shelf.

A Nested Loop Finishes the Inner Loop Every Outer Pass

OUTER

Choose one shelf.

INNER

Check every box on that shelf.

A Nested Loop Finishes the Inner Loop Every Outer Pass

OUTER

Choose one shelf.

INNER

Check every box on that shelf.

COUNT

Multiply outer × inner.

Finish every box on shelf 1 before moving to shelf 2.

Nesting Conditionals

Loops can nest inside loops. Conditionals can nest inside conditionals the same way.

REAL-WORLD SCENARIO

A Food Bank Script Approved a Negative Donation

```
# donations.txt
ADD canned_beans 40
ADD rice -12
REMOVE pasta 10
ADD diapers 25
```

Illustrative data

The script checked each row's action
It never checked the amount inside ADD
“ADD rice -12” cut stock by 12

One check passed, so the bad row got in.
Where does the second check belong?

Illustrative scenario, not a real event

A Nested Conditional Checks in Layers

1 OUTER IF

Pick the category.

A Nested Conditional Checks in Layers

1 OUTER IF

Pick the category.

2 INNER IF

Check the detail.

A Nested Conditional Checks in Layers

1 OUTER IF

Pick the category.

2 INNER IF

Check the detail.

RULE

Inner check runs only if outer matched.

Closet analogy: choose shirts, pants, or jackets first — then choose where inside that section.

CODE ALONG

Open the Code-Along

Open `Module1_Lesson2_nested-loops-conditionals_STUDENT.py`
Run it after every section: `python3` + the file name

Sections, in order:

1. Nested Loop, Live · the inner loop finishes first
2. Nested Conditional, Live · the inner check needs the outer match
3. Putting It Together: Loop + Nested Conditional · the project's shape

Nested Loop, Live

```python
for row in range(2):
    for col in range(3):
        print(row, col)
```

Nested Conditional, Live

python

```python
operation = "ADD"
quantity = -5

if operation == "ADD":
    if quantity > 0:
        print("Valid add")
    else:
        print("Invalid quantity")
elif operation == "REMOVE":
    print("Processing remove")
```

Putting It Together: Loop + Nested Conditional

python

```python
transactions = ["ADD 10", "REMOVE 3", "ADD -5"]
for line in transactions:
    operation, amount = line.split()
    amount = int(amount)
    if operation == "ADD":
        if amount > 0:
            print(line, "- valid")
        else:
            print(line, "- invalid quantity")
    elif operation == "REMOVE":
        print(line, "- processing remove")
```

CODE-ALONG DONE

Back to the Slides: Review

You just typed it and ran it. Now check that it stuck, without the code in front of you.

Watch It Run: Nested Loops

Step 1 of 6

row is 0, col is 0: the inner loop starts

```
for row in range(2):
    for col in range(3):
        print(row, col)
```

OUTPUT

```
0 0
```

ANIMATED IN CLASS: watch it run in the lecture recording

ADD TO THE BOARD

Put the Print Statements in Execution Order

```
for row in range(2):
    for col in range(2):
        print(row, col)
```

A

(1, 0)

B

(0, 0)

C

(1, 1)

D

(0, 1)

QUICK CHALLENGE

3 outer passes × 2 inner passes: how many lines print?

A

5

B

6

C

9

ANSWER KEY

Nested Loop Trace

Correct answer: 6

3 outer passes × 2 inner passes each = 6 total prints. Multiply the counts; don’t add them.

QUICK CHALLENGE

A nested loop should print 6 lines but prints only 3. Most likely cause?

A The inner line is outside the outer loop body

B Python randomly skips iterations

C `range()` only works on outer loops

ANSWER KEY

Nested Loop Debug

Correct: The inner line is outside the outer loop body

Indentation is the nesting. If the line isn’t inside the inner loop, it runs once per outer pass instead of once per inner pass.

QUICK CHALLENGE

The inner conditional never runs. What should you trace first?

A. A missing import

B. Whether Python skipped it randomly

C. The outer condition

ANSWER KEY

Nested Conditional Debug

Correct: The outer condition

An inner `if` runs only after the outer branch containing it is taken. If the inner check never fires, trace the outer condition first.

CHECK FOR UNDERSTANDING

In the Putting It Together code, which check flags `"ADD -5"`?

A

```
operation == "ADD"
```

B

```
amount > 0
```

C

```
operation == "REMOVE"
```

ANSWER KEY

Which Check Flags `ADD -5`?

Correct answer: B, `amount > 0`

`operation == "ADD"` is true, so Python enters that branch. Then `amount > 0` is false for `-5`, so the inner `else` prints `"invalid quantity"`. The `REMOVE` check never runs.

TIP

“How Would You Trace a Nested Loop?”

BY HAND

Write every iteration in order.

PYTHON TUTOR

Step through execution line by line.

Pick one. Answer with PREP: point, reason, example, point.

MODEL ANSWER

Model Answer: PREP

POINT

I'd trace it by hand first.

REASON

It builds the instinct a timed screen tests, with no tools.

EXAMPLE

Today's 2-by-3 loop is only six `(row, col)` lines to write.

POINT, RESTATED

By hand first, then Python Tutor to confirm a tricky one.

So Far, We've Covered…

BEFORE

Loops and conditionals

TODAY

Nest either one, one level deeper

NEXT

Functions, parameters, and scope

KEEP LEARNING

Python Tutor — Visualize Nested Loops

PRACTICE • 5–10 MIN

Step through a nested loop line by line.

Open Python Tutor

Python Docs — for Statements